



MRAC / MAAi

Hardware , Term III 25/26

Hardware III

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Guided Comparative Assembly.

An interactive construction kitchen — configure, build, compare.

Masonry

brick by brick

3D Printed

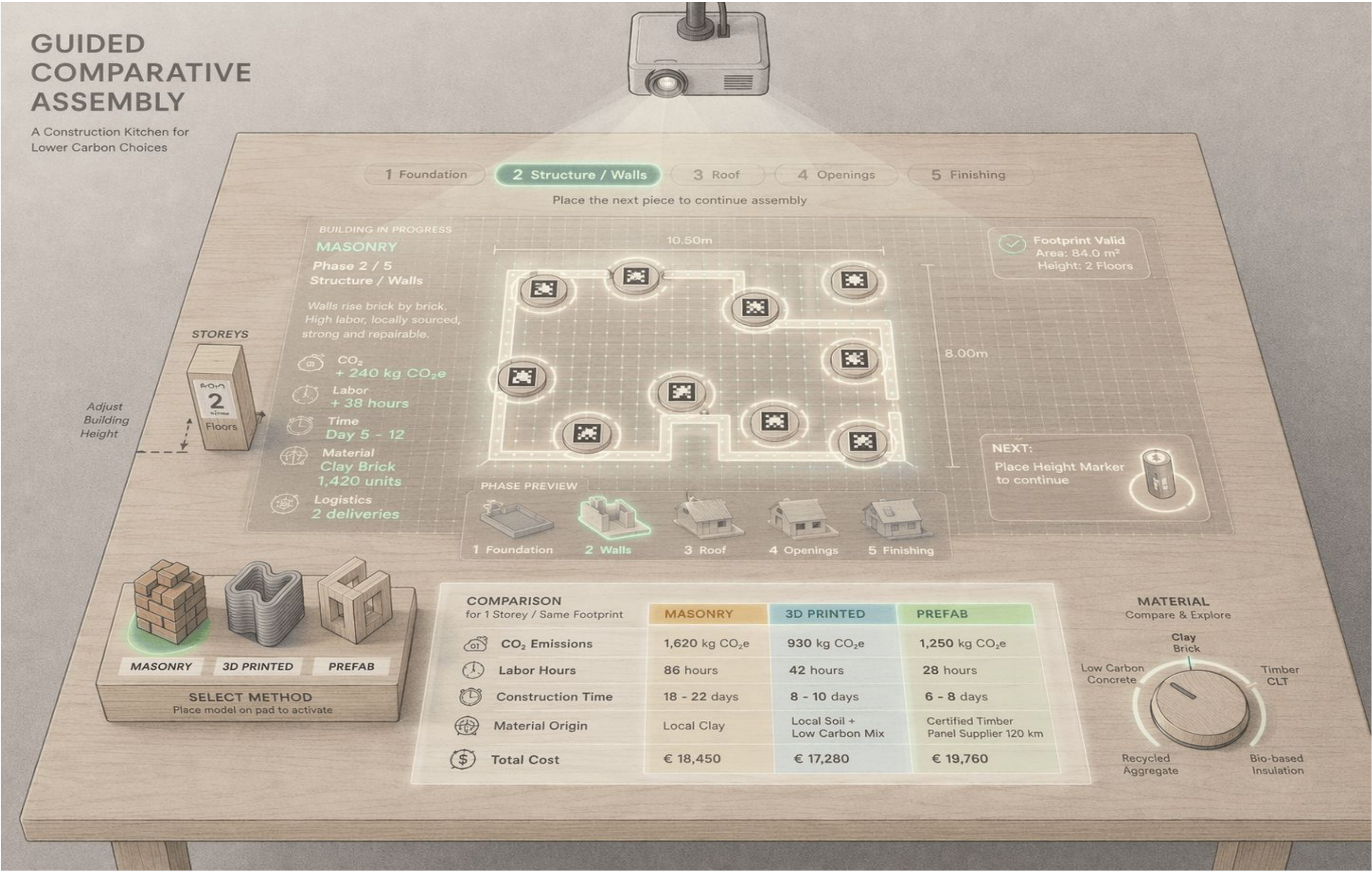
layer by layer

Prefab

module by module

The Concept

A hands-on exhibit for children, young architects, and decision makers.



USER FLOW

No screens, no buttons.

01 Place model on RFID pedestal

02 Arrange 10 pucks = footprint

03 Rotate dials — height + material

04 System validates feasibility

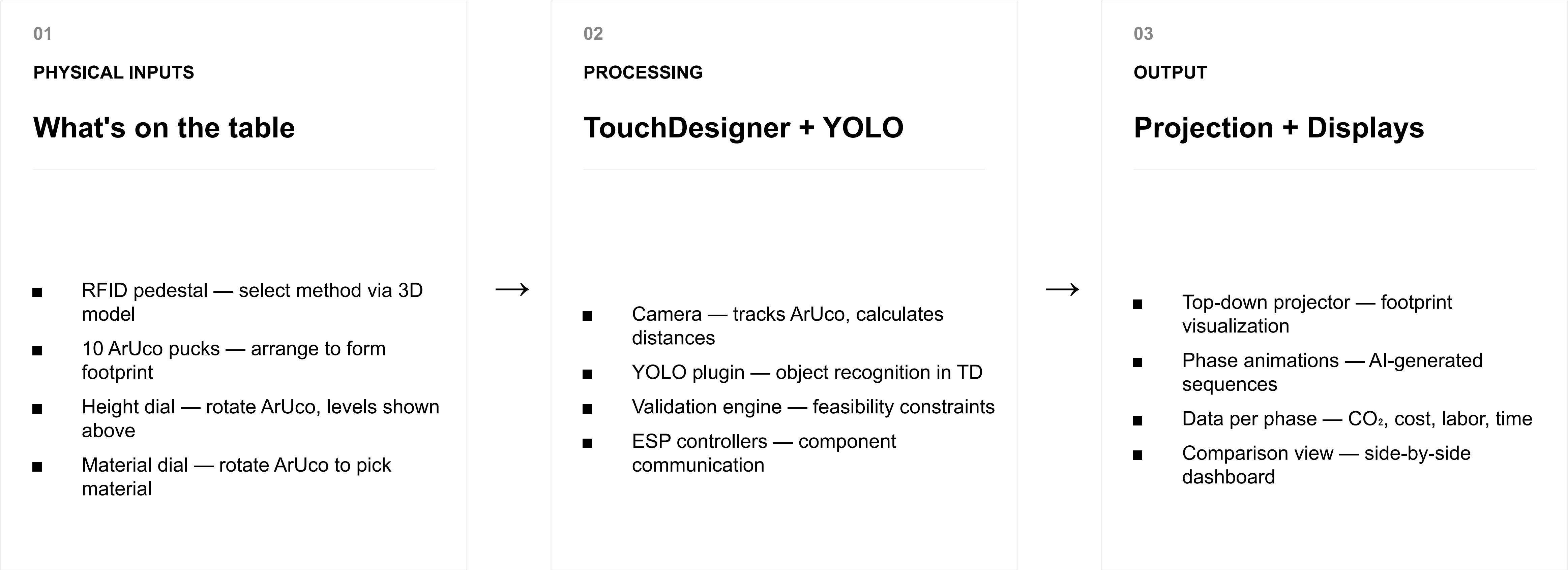
05 5 building phases animate

06 Save & compare builds

FIG. 01 · GUIDED COMPARATIVE ASSEMBLY

System Architecture

Physical components → Processing → Projection feedback



Pipeline First

Before anything else — the software + hardware pipeline has to work.

01

HARDWARE

What we need on day one

- Laptop (GPU preferred)
 - Short-throw projector (3000 lm)
 - Webcam (1080p · wide angle)
 - ESP32-WROOM-32
 - MFRC522 RFID reader
 - 3× RFID coin tags (MIFARE)
 - 12× ArUco markers (Bambu X1C)
- ≈ 25 – 40 € extra material*

02

SOFTWARE STACK

Runtime + firmware

- TouchDesigner 2023.11+
 - OpenCV · opencv-contrib-python
 - Arduino IDE + ESP32 core
 - MFRC522 library
 - ArduinoOSC (hideakitai)
 - Python 3.11 (scripting)
- free non-commercial license*

03

DATA FLOW

How signals travel

RFID tag
ESP32 → WiFi / OSC
→ TouchDesigner

ArUco
Webcam → OpenCV
→ TouchDesigner

TD
Homography warp
→ Projector

12 ARUCO MARKERS · DICT_4X4_50

IDs 0 – 9

Footprint pucks
user-arrangeable

ID 10

Story dial
rotate for levels

ID 11

Material dial
rotate for variant

What We Track

Parameters measured per phase · Development priorities & research

PARAMETERS PER PHASE

CO₂ emissions per ton · per phase	€ total cost broken down by phase
hrs labor hours worker time · safety	kg material usage concrete, wood, cement
km logistics transport · Catalonia	days construction duration per phase

DEVELOPMENT PRIORITIES

P1	Pipeline live (first) TD + webcam + ArUco + OSC — full data path working
P2	RFID + ESP Pedestal integration, method selection, ESP comms
P3	Building phases AI-generated animations, per-phase info display
P4	Data + Comparison Cost / CO₂ tracking, dataset save, comparison view

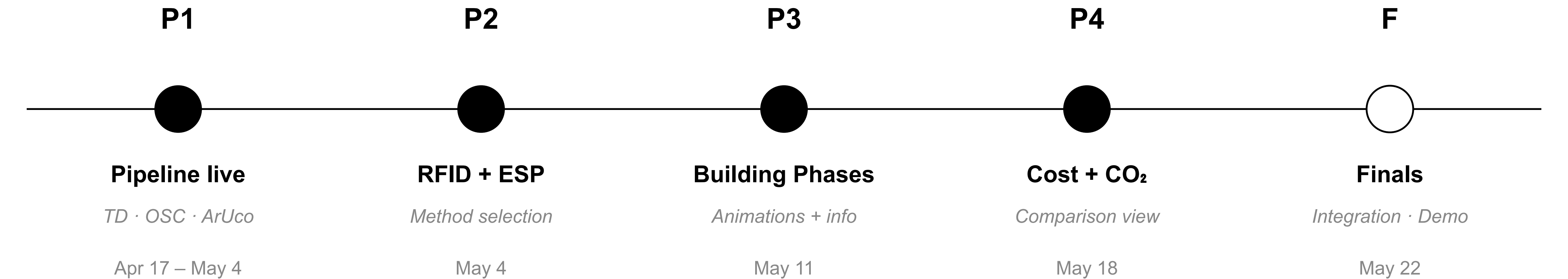
5 BUILDING PHASES · DISPLAYED PER METHOD

1 Foundation	2 Structure	3 Roof	4 Openings	5 Finishing
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RESEARCH10+ companies per method · 3D-printed ArUco pucks · YOLO plugin for TD · Catalonia logistics data

Roadmap to Finals

Five milestones, four weeks · April 17 → May 22, 2026



TEAM

Leo · Elais · Rafik · Seid · Onur · Nithik

Instructors: Hamid Peiro · Aleksandra Kraeva

IAAC MRAC + MAAI · April – May 2026

INSPIRED BY

Swiss Museum DJ Table

ArUco + projection workflow

Stefana Parascho · Gramazio Kohler

ETH robotic fabrication research

Fologram

Mixed-reality construction guidance